

Faculty Management System

Java Swing · JDBC · MySQL · MVC Architecture

FMS-Group-O.pdf

Team & Individual Contribution Report

GitHub Organization: FMS - Group O

Repository: <https://github.com/thayassan/Faculty-Management-System>

Team Members

No.	Student Number	Full Name	Module Owned
1	CT/2022/027	Thayassan S.	Project Lead
2	CT/2022/008	Dinushan R.	Student Module
3	CT/2022/022	Pradeep E.	Lecturer Module
4	CT/2022/029	Karushman S.	Course, Enrollment
5	CT/2022/040	Sandeepa K.H.	Admin, Department
6	CT/2022/059	Kathiravel S.	Database Design

Project Overview

The Faculty Management System (FMS) is a desktop application built with Java Swing that manages students, lecturers, courses, departments and degrees for the Faculty of Computing & Technology. The application follows the MVC design pattern (model / view / controller / dao / util / main) and persists data through JDBC to a MySQL database. It supports three roles — Admin, Student and Lecturer — each with a dedicated dashboard, and the Admin role has full CRUD access to every entity.

Individual Contributions

Work was divided by functional module so that each member owns a complete vertical slice of the MVC architecture (model, DAO, controller and, where applicable, view) for their assigned entity. This allows each member to independently build, test, and explain their components during the code walkthrough.

1. Thayassan S. (CT/2022/027)

Module: Project Lead & Core / Authentication Module

Set up the MVC package structure and application entry point (Main.java); implemented the role-based Sign In / Sign Up screen (Admin, Student, Lecturer) and its controller; built the User model and UserDAO for credential validation and role lookup; implemented the JDBC connection utility (DBConnection/DBUtil) used by every other DAO; coordinated repository setup, branch merges and final integration/build testing across all modules.

Files owned:

- `src/com/faculty/main/Main.java`
- `src/com/faculty/view/LoginView.java`
- `src/com/faculty/controller/LoginController.java`
- `src/com/faculty/model/User.java`
- `src/com/faculty/dao/UserDAO.java`
- `src/com/faculty/util/DBConnection.java`
- `src/com/faculty/util/DBUtil.java`

2. Dinushan R. (CT/2022/008)

Module: Student Module

Implemented the Student entity model and StudentDAO (CRUD + joined queries for timetable/enrollment); built the student dashboard view with Profile Details, Timetable and Courses Enrolled tabs; wired profile editing (save changes) through StudentController.

Files owned:

- `src/com/faculty/model/Student.java`
- `src/com/faculty/dao/StudentDAO.java`
- `src/com/faculty/controller/StudentController.java`
- `src/com/faculty/view/StudentDashboardView.java`

3. Pradeep E. (CT/2022/022)

Module: Lecturer Module

Implemented the Lecturer entity model and LecturerDAO (profile CRUD, department linkage); built the Lecturer dashboard view showing profile details, weekly timetable and assigned teaching courses; connected view actions to LecturerController.

Files owned:

- `src/com/faculty/model/Lecturer.java`
- `src/com/faculty/dao/LecturerDAO.java`
- `src/com/faculty/controller/LecturerController.java`
- `src/com/faculty/view/LecturerDashboardView.java`

4. Karushman S. (CT/2022/029)

Module: Course, Enrollment & Timetable Module

Implemented Course, Enrollment and TimetableEntry models with their DAOs; built the CRUD logic used by the Admin > Courses tab (add/edit/delete, assign lecturer); implemented enrollment/grade record queries and the timetable lookup logic consumed by the Student and Lecturer dashboards.

Files owned:

- `src/com/faculty/model/Course.java`
- `src/com/faculty/dao/CourseDAO.java`
- `src/com/faculty/model/Enrollment.java`
- `src/com/faculty/dao/EnrollmentDAO.java`
- `src/com/faculty/model/TimetableEntry.java`
- `src/com/faculty/dao/TimetableDAO.java`

5. Sandeepa K.H. (CT/2022/040)**Module:** Admin, Department & Degree Module

Built the Admin dashboard shell (sidebar navigation across Students / Lecturers / Courses / Departments / Degrees, Add / Edit / Delete / Save changes actions) and AdminController; implemented Department and Degree models and DAOs, including HOD/staff-count and degree-to-department linkage logic used in the Departments and Degrees tabs.

Files owned:

- `src/com/faculty/controller/AdminController.java`
- `src/com/faculty/view/AdminDashboardView.java`
- `src/com/faculty/model/Department.java`
- `src/com/faculty/dao/DepartmentDAO.java`
- `src/com/faculty/model/Degree.java`
- `src/com/faculty/dao/DegreeDAO.java`

6. Kathiravel S. (CT/2022/059)**Module:** Database Design, UI Styling, QA & Documentation

Designed the relational schema (tables, primary/foreign keys, constraints) and wrote the sample data seed script; implemented the shared UIStyle theme (deep purple/indigo palette, rounded buttons, table styling) reused across every view; carried out end-to-end testing of all three roles, wrote the README, and recorded/edited the demo video and compiled this documentation.

Files owned:

- `database/schema.sql`
- `database/sample_data.sql`
- `src/com/faculty/view/UIStyle.java`
- `README.md`
- `docs/` (this report)
- `demo/` (demo video)

Collaboration & Git Workflow

Each member was added as a collaborator on the shared repository and pushed their own module from their own GitHub account, so the commit history reflects genuine individual authorship (as required for individual grading). The workflow used:

- Clone the repository and create a personal feature branch named `feature/<module>-<name>`, e.g. `feature/student-module-dinushan`.
- Implement and test the assigned model / DAO / controller / view classes locally against the shared MySQL schema.
- Commit in small, descriptive steps (e.g. one commit per class) rather than a single bulk commit.

- Push the feature branch and open a Pull Request into main.
- The project lead reviews and merges each PR after a short walkthrough with the author.

This ensures the GitHub commit graph shows six distinct contributors, each traceable to the module described above, satisfying the assignment's individual-grading requirement.